Exercise list — Topic 4: Moral hazard, credit rationing, asset substitution, debt overhang

Answer key (full derivations in `lista-topic4-moral-hazard-overhang-solutions.pdf`)

1. DIP financing and existing creditors

Debt overhang blocks junior financing: in Grinblatt–Titman's Lily example a junior lender gets at most $150 - 110 = 40 < 100$, so the project (NPV 50) is passed up and the old lenders get 0. A senior DIP loan of 100 is riskless and leaves the old lenders $\min\{150 - 100, 110\} = 50 > 0$. In general, with a safe DIP loan $\min\{V_s + X_s - I, F\} \geq \min\{V_s, F\}$ in every state: they lose priority but the pool grows by more. Lecture numbers (500/200, $F = 300$, $I = 60$, +100): D rises from 250 to 270 and E from 100 to 120; $20 + 20 = 40 =$ NPV. Caveat: DIP can also keep alive firms that should be liquidated (Result 16.11).

2. Asset substitution: project A or B

(a) $NPV_A = 0.5(120) + 0.5(60) - 100 = -10$; $NPV_B = 101 - 100 = 1$.
(b) Equity gets $\max\{X - 100, 0\}$: $E_A = 0.5(20) + 0.5(0) = 10$, $E_B = 0.5(1) + 0.5(1) = 1$; debt $D_A = 80$, $D_B = 100$. Equity wants **A**: it loses 11 of firm value but takes 20 from the creditors ($E_A - E_B = -11 + 20 = 9$). Levered equity is a call option and prefers risk.

3. Debt overhang on vacant land

(a) $E = 0$, $D = 10/1.10 = 9.09$. **(b)** $NPV = -20 + (35 - 10)/1.10 = 2.73$ (the forgone vacant-land value counts). **(c)** $E = (35 - 15)/1.10 = 18.18$, $D = 15/1.10 = 13.64$. **(d)** No: equity pays 20 for 18.18 (-1.82); the debt gains $13.64 - 9.09 = 4.55$; $-1.82 + 4.55 = 2.73 =$ NPV. The creditors capture more than the whole NPV: debt overhang.

4. Short-term debt and incentive problems

Conflicts arise because equity changes the value of a claim whose terms are fixed. Debt repriced at the time of the decision sets F_j so that $\mathbb{E}[\min\{X_j, F_j\}] = L$, hence $E_j = V_j - L$: equity internalizes the whole value and chooses the efficient strategy. In Exercise 2, $E_A = 90 - 100 < 0$, $E_B = 101 - 100 = 1$, so B is chosen. Debt overhang (Myers): debt maturing before the new project is repriced with it, so the benefit of a safer firm shows up as a lower rate for equity. Asset substitution: the rate is reset after the risk choice. Cost: rollover and interest-rate risk.

5. Random financing

(i) $x\, p_H(R - R_b) \geq xI - \hat{A}$; with (IC$_b$) $R_b \geq B/\Delta p$ (shirking gives the borrower $x(p_L R + B - I) < 0$) this becomes $x\bar{A} \leq \hat{A}$.
(ii) With breakeven binding, $U_b = x(p_H R - I)$, increasing in x; $x \leq \hat{A}/\bar{A}$ rises with $\hat{A}$, so $\hat{A} = A$, $x^* = A/\bar{A}$, $U_b^* = (A/\bar{A})(p_H R - I)$ and $R_b = B/\Delta p$. x^* rises linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$ (then stays at 1): utility no longer jumps at $\bar{A}$. Example: $\bar{A} = 40$, $A = 10$, NPV 20: $x^* = 0.25$, $U_b^* = 5$. Matches Tirole p. 571.

6. Zymase: leverage and strategy choice

(a) $\mathbb{E}[X]$: A 75, B 70, C $0.1(300) + 0.9(40) = 66$: **A**.
(b) $F = 40$: equity A 35, B $0.5(100) = 50$, C $0.1(260) = 26$: **B**.

(c) $F = 110$: equity A 0, B $0.5(30) = 15$, C $0.1(190) = 19$: **C**.
(d) Agency cost $= \mathbb{E}[X_A] - \mathbb{E}[X_{\text{chosen}}]$: $75 - 70 = 5$ with \$40m of debt, $75 - 66 = 9$ with \$110m (payoff-date expected values; divide by $1 + r_f$ for today).

7. Ajax Manufacturing: debt overhang

16.12 $NPV = 0.2(85) + 0.8(150) - 75 = 137 - 75 = 62$. Equity is junior to the \$70m bond: it gets 15 or 80, worth $0.2(15) + 0.8(80) = 67 < 75$. Even all the equity cannot raise \$75m: **no**. (Bond would be worth 70; $70 - 8 = 62$.)
16.13 Same NPV, $0.2(5) + 0.8(170) - 75 = 62$, but equity gets 0 or 100, worth $80 \geq 75$: **yes** (new holders take $75/80 = 93.75\%$; old keep 5). The bond falls to $0.2(5) + 0.8(70) = 57$: the riskier process moves 13 from the bond to equity, which cures the overhang; the bondholders still prefer it to getting 0.

8. Project riskiness and credit rationing

Same minimum stake $B/\Delta p$; $\bar{A}^j = p_H^j B/\Delta p - (p_H^j R^j - I)$ and equal NPVs give $\bar{A}^A - \bar{A}^B = (p_H^A - p_H^B)B/\Delta p > 0$. The riskier **variant B is less prone to credit rationing**: the safe project pays the success-contingent rent more often, so less income is pledgeable. Example ($I = 100$, $\Delta p = 0.2$, $B = 9$, $p_H^A = 0.8$, $p_H^B = 0.6$, $p_H R = 120$): $\bar{A}^A = 16$, $\bar{A}^B = 7$. Matches Tirole p. 578; the book adds that if a misbehaviour contract can finance the safe project ($I - A \leq p_L^A R^A$), the risky one may be the more rationed.

9. Entrepreneurial risk aversion (Tirole 3.2)

(i) $p_H < 1$: failure occurs on path, so $c^F \geq c_0$; cheapest IC contract $c^F = c_0$, $c^S = c_0 + B/\Delta p$. Pledgeable income $p_H(R - B/\Delta p) - c_0$, so $\bar{A} = I + c_0 - p_H(R - B/\Delta p) = \bar{A}_{rn} + c_0 > c_0$ (rationing on $[c_0, \bar{A})$). $p_H = 1$: failure only after shirking, punished by $-\infty$; IC is free, $\mathcal{P} = R - c_0$, $\bar{A} = I + c_0 - R < c_0$: everyone is financed (needs $c_0 > 0$ under limited liability).
(ii) $\max p_H u(c^S) + (1-p_H)u(c^F)$ s.t. $p_H(R - c^S) - (1-p_H)c^F \geq I - A$, $(\Delta p)[u(c^S) - u(c^F)] \geq B$, $[c^F \geq 0]$, $V \geq u(A)$. Both IR_l and IC bind ($\mu = 0$ would mean full insurance); $1/u'(c^S) = (1 + \mu\Delta p/p_H)/\lambda$, $1/u'(c^F) = (1 - \mu\Delta p/(1 - p_H))/\lambda$. No LL: financed iff the risk premium of the incentive lottery $\leq NPV$. LL: need $A \geq A_{LL} = I - p_H[R - u^{-1}(u(0) + B/\Delta p)]$ plus $V(A) \geq u(A)$; above A_{LL}, $c^F > 0$ (partial insurance).

10. Maximal incentives principle (Tirole 3.18)

IR_l binds, so $U_b = NPV$ for every feasible contract. Max pledgeable income at $R_b^F = 0$, $R_b^S = B/\Delta p$: financed iff $p_H(R - B/\Delta p) \geq I - (A + R^F)$. Debt D with $p_H D + (1 - p_H)R^F = I - A$ has $R^F < D \leq R^S - B/\Delta p$, so it is optimal. Optimal set: all $R_b^F \in [0, A - \bar{A}]$ with IR_l binding, $\bar{A} = p_H B/\Delta p - NPV$: unique (debt) only at $A = \bar{A}$. With variable investment $A = \bar{A}(I)$ always binds, so debt is unique.

11. Retained-earnings benefit (Tirole 3.11)

(i) Project 2 financed iff $B^2 \leq \hat{B}^2(R) = \frac{\Delta p^2}{p_H^2}(NPV^2 + R)$; never with $R = 0$. Gross utility $W(R) = R + F(\hat{B}^2(R))NPV^2$; shadow value $\mu = 1 + f(\hat{B}^2)\frac{\Delta p^2}{p_H^2}NPV^2 \geq 1$.
(ii) IC^1: $(\Delta p^1)[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2] \geq B^1$, so the minimum stake $\tilde{R}_b^1 < B^1/\Delta p^1$ and pledgeable income $p_H^1(R^1 - \tilde{R}_b^1)$ is larger than stand-alone. Example ($B^2 \sim U[7, 9]$, $NPV^2 = 10$; $B^1/\Delta p^1 = 2$):

$\mu = 13/3$, $\tilde{R}_b^1 = 11/13$, pledgeable $119/26 \approx 4.576923 > I^1 = 4.5$; threshold falls from 0.5 to -0.076923.

12. Renegotiation and debt forgiveness (Tirole 3.6)

(i) Status quo: borrower $p_L R_b + BI$, lender $p_L(RI - R_b)$. Gains need $\hat{R}_b \geq BI/\Delta p$; best for the lender $\hat{R}_b = BI/\Delta p$, so renegotiation iff $(\Delta p)RI - p_H BI/\Delta p + p_L R_b \geq 0$.
(ii) Fictitious project: income $(\Delta p)RI$, investment 0, cash $p_L R_b$; the condition is "pledgeable income $\geq 0-$cash".
(iii) Lender gets $p_L(RI - R_b)$; capacity $I \leq kA$ if $p_H B/\Delta p \geq (\Delta p)R$, and $A/(1 - p_L R) < kA$ otherwise: never above kA.
(iv) Any renegotiated outcome is an IC contract lenders accept ex ante, so it could be signed directly; without renegotiation the NPV is negative. WLOG $R_b \geq BI/\Delta p$.

13. Loan size and collateral (Tirole 4.17)

$\max p_H R(I) - I - (1 - p_H)[C - \phi(C)]$ s.t. $p_H[R(I) - BI/\Delta p + C] + (1 - p_H)\phi(C) \geq I - A$. FOCs ($\theta = \mu/(1 + \mu) > 0$): $p_H R'(I) - 1 = \theta\, p_H B/\Delta p$, $\phi'(C) = 1 - \theta/(1 - p_H)$. $d\theta/dA < 0$, so as A falls: $I \downarrow$, $C \uparrow$. Agency cost $b = p_H B/\Delta p$: $dI/db < 0$ always; $dC/db > 0$ iff $\varepsilon(I)(1 + \theta b) > \theta(1 - \theta)b$, $\varepsilon = -IR''/R'$. Fails e.g. for $R = 3I^{0.95}$, $\phi = \ln(1 + C)$, $p_H = 0.5$, $A = 1$: $b = 1 \to C = 1.9003$, $b = 4 \to C = 0.3292$.

14. Dealing with asset substitution (Tirole 7.3)

(i) LHS of (1) = maximal pledgeable income; $U_b^* = (1 - x)L_0 + x p_H R - I$; optimal contracts: $(1 - x)(L_0 - r_b) + x p_H(R - R_b) = I - A$, $R_b \geq B/\Delta p$; at $\bar{A}$: $r_b = 0$, debt $D = R - B/\Delta p$.
(ii) (2) = FOC of the NPV in L: L_0 is efficient. Step contract; no substitution iff $(1 - x)r_b \geq x\tau(0)R_b$. Min rent: $R_b = B/\Delta p$, $r_b = \frac{x\tau(0)}{1-x}\frac{B}{\Delta p}$, so $A^* = \bar{A} + x\tau(0)B/\Delta p$. Example: $\bar{A} = 0.7$, $r_b = 0.375$, $A^* = 0.85$.

15. Borrower-friendly bankruptcy court (Tirole 4.9)

(i) Pledge r and L; $R_b = B/\Delta p$. (1): continuation efficient; (2): always continuing is not fundable, always liquidating is.
(ii) $\max E[r] + LF(r^*) + \int_{r^*} \rho_1 f - I$ s.t. $E[r] + LF(r^*) + \int_{r^*}^{\bar{r}} \rho_0 f = I - A$; r^* = smallest root, $\rho_0(r^*) < L$.
(iii) Short-term debt $d = r^*$; default = bankruptcy = liquidation; $dr^*/dA = -1/[(L - \rho_0(r^*))f(r^*)] < 0$.
(iv) $r^* > \hat{r}$: investors' income $\leq \max\{L/2, \rho_0\}$, strictly below breakeven: no financing.
(v) $r^* < \hat{r}$: either no financing or cutoff $r^{**} \in (r^*, \hat{r}]$; lenders break even, borrower's NPV falls. Example: $I - A = 3.31$, $r^* = 0.0040$, $r^{**} = 0.0471$.

16. Default as insurance: the default cutoff (original)

(a) Cash-flow default iff $A + Y < D$, i.e. $Y < 7$; probability $F(7) = 0.1$.
(b) Repay: $u = \sqrt{Y - 7}$; default: $u(9) - 1 = 2 > 0$. Strategic default iff $\sqrt{Y - 7} < 2$; $Y^* = 7 + 4 = 11$, and $7 < 11 < 16 = D + E$ (garnishment $Y - E < 7 < D$, so the cap never binds).
(c) $p = F(11) = 6/20 = 0.3$: 0.1 cash-flow, 0.2 strategic.
(d) $Y = 6$: cash-flow default, $C = 9$, $u = 2$, creditors 2. $Y = 9$: could repay ($u = \sqrt{2} = 1.414 < 2$) but defaults, creditors 5. $Y = 20$: repays, $C = 13$, $u = 3.606$, creditors 12.

(e) $E[Y - E \mid Y < 11] = 8 - 4 = 4$; consumption floor $A + E = 9$: default swaps low, risky consumption for a fixed floor.

17. Default as insurance: comparative statics, pricing, ex ante welfare (original)

(a) From $u(A + Y^* - D) = u(A + E) - \sigma$ and $c_r < c_d$ (since $\sigma > 0$): $dY^*/dD = 1$, $dY^*/dA = u'(c_d)/u'(c_r) - 1 < 0$, $dY^*/d\sigma = -1/u'(c_r) < 0$, $dY^*/dE = u'(c_d)/u'(c_r) > 0$. Baseline ($c_r = 4$, $c_d = 9$, ratio 2/3): 1, $-1/3$, -4, 2/3; $dp/dx = \frac{1}{20}dY^*/dx$: 0.05, $-1/60$, -0.2, 1/30 (slide 44).
(b) $Q = 0.7(12) + \frac{1}{20}\int_5^{11}(y-4)dy = 8.4 + 1.2 = 9.6$; rate $12/9.6 - 1 = 25\%$.
(c) Regime N: $D_N = 9.6$, $EU_N = \frac{1}{30}[20.4^{3/2} - 0.4^{3/2}] = 3.062880$, $CE = 9.381$. Regime B ($\sigma = 1$): $D_B = 12$, $EU_B = 0.6 + \frac{1}{30}[18^{3/2} - 8] = 2.878918$, $CE = 8.288$. $E[C] = 10.4$ in both. N preferred.
(d) $Y^* = D - A + c_r$, $40Q = -D^2 + 42D + (c_r - 9)^2 - 1$. $\sigma = 0.1$: $c_r = 8.41$, $D_B = 21 - \sqrt{56.3481} = 13.4935$, $Y^* = 16.9035$, $p = 0.5952$, rate 40.56%, $EU_B = 3.148474$, $CE = 9.913$: B preferred.
(e) Same mean consumption, single crossing ⇒ insurance gain $E[u(C_B)] - EU_N > 0$; welfare gap = gain $-\sigma p$: $0.116 - 0.3 < 0$ at $\sigma = 1$, $0.145 - 0.060 > 0$ at $\sigma = 0.1$. $\bar{\sigma} \approx 0.26$ (0.2648).

18. Karlan–Zinman: selection vs moral hazard (original)

(a) H iff $\Delta\pi_\theta[Y(\theta) - (1 + r_c) + C_b] \geq c$. S: gains 0.03–0.07 < 0.17, always L. R: 0.19, 0.23, 0.15, 0.19 for $(10\%, 0.25)$, $(10\%, 0.45)$, $(30\%, 0.25)$, $(30\%, 0.45)$: L only at $(30\%, 0.25)$.
(b) Expecting $r_c = r_o$, $C_b = 0.25$: S gets 0.15 at 10%, -0.01 at 30%; R gets 0.245 and 0.125. Take-up 100% at 10%, 50% (only R) at 30%.
(c) Default rates (no / yes dynamic incentive): A 25%/25%; B 30%/30%; C 50%/30%.
(d) Selection: B−A = +5 pp. Burden: C−B = +20 pp (no incentive), 0 (with). Dynamic: C −20 pp; 0 in A, B (ICs not near binding there).
(e) C−A = 25 pp = 5 + 20; with $r_o = r_c$ always, the parts cannot be separated.
(f) $1 + r_c \leq Y_R + C_b - c/\Delta\pi_R = 0.95 + C_b$: $r_c \leq 20\%$ without, 40% with the incentive; HT's $(\Delta p)(R_b + C_b) \geq B$; pledgeable $0.84 \to 0.98$.

19. Reading a three-rate experiment (original; hypothetical data)

(a) Cells: A 11.5/8.5%, B 17.6/14.4%, C 18.8/15.2%; groups 10, 16, 17%; pooled 16.29% vs 13.0%. Selection B−A = 6.0 pp; burden C−B = 1.0 pp; dynamic −3.29 pp (−3.0, −3.2, −3.6 within).
(b) SE 0.01283, 0.01355, 0.01090; $z = 4.68$, 0.74, -3.01: selection and dynamic significant at 5%, burden not.
(c) Saturated in groups, C_i balanced (FWL): $\beta_o = 6/10 = 0.6$, $\beta_c = 1/10 = 0.1$, $\beta_b = -3.29$, $\alpha = -9.36$.
(d) $\mathcal{R}(30\%) = 0.90(1.30) = 1.170 > \mathcal{R}(40\%) = 0.83(1.40) = 1.162$; $-0.008 = +0.090$ (rate) -0.084 (selection) -0.014 (burden): backward-bending supply, ration rather than raise the rate (Stiglitz–Weiss, lemons).
(e) IC $(\Delta p)(R_b + C_b) \geq B$; $\mathcal{P} = 0.9(1.5 - 0.5 + C_b)$: 0.9 ($\bar{A} = 0.1$) for $C_b = 0$, 1.08 ($\bar{A} = -0.08$, no rationing) for $C_b = 0.2$; utility $0.35 - 0.02 = 0.33$.
(f) Cannot separate selection on risk type from selection on anticipated effort (slide 47).